

MSc Big Data & Finance

Master Thesis — Preliminary Work Submission

Machine Learning to Identify ESG Rating Inconsistencies and Improve Portfolio Outcomes

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1. Finalized Research Question

This thesis investigates whether machine learning (ML) techniques can systematically identify inconsistencies between Environmental, Social, and Governance (ESG) ratings and underlying firm fundamentals, and whether penalizing ESG scores for such inconsistencies improves risk-adjusted portfolio outcomes.

The central research question is:

Can machine learning identify inconsistencies between ESG ratings and firm fundamentals, and does penalizing ESG scores for such inconsistencies improve risk-adjusted portfolio outcomes relative to standard ESG integration?

This question builds directly on the February 2026 submission, which was approved by the academic supervisor without modification. The research is motivated by the growing evidence that ESG ratings from major providers diverge substantially from one another (Berg, Kölbel, and Rigobon, 2022) and that this divergence may be driven less by genuine differences in firm sustainability performance than by methodological choices, disclosure incentives, and appearance-driven reporting (Christensen, Hail, and Leuz, 2021). The thesis proposes that machine learning offers a disciplined, data-driven approach to disentangle the informational content of ESG ratings from their noise component.

2. Research Sub-Questions

The central research question is decomposed into four operational sub-questions, each corresponding to a distinct empirical step in the methodology:

RQ1 — Predictive Power of Fundamentals:

To what extent can machine learning models accurately predict ESG ratings from observable firm-level fundamentals, and which fundamental variables are most predictive?

RQ2 — Nature of ESG Inconsistency:

What firm characteristics are systematically associated with a large gap between observed ESG ratings and ML-predicted ESG ratings, and is this gap related to known proxies for greenwashing or disclosure opacity?

RQ3 — Portfolio Performance:

Do ESG-tilted portfolios that penalize high-inconsistency firms achieve superior risk-adjusted returns — measured by Sharpe ratio, drawdown, and tracking error — compared to portfolios using raw ESG scores?

RQ4 — Signal Purity:

Does the inconsistency-adjusted ESG score exhibit lower cross-provider rating disagreement than the raw ESG score, suggesting it captures a cleaner sustainability signal?

3. Hypotheses

Three primary hypotheses and one auxiliary hypothesis are formulated. Each hypothesis is stated in its null and alternative forms to allow for rigorous empirical testing.

Hypothesis 1 — ML Predictive Accuracy

H1₀: Machine learning models do not predict ESG ratings from firm fundamentals with statistically significant accuracy (out-of-sample $R^2 \leq 0$).

H1₁: Machine learning models predict ESG ratings from firm fundamentals with statistically significant accuracy (out-of-sample $R^2 > 0$), and non-linear models (Random Forest, XGBoost) outperform the linear Ridge regression baseline.

Rationale: Gu, Kelly, and Xiu (2020) demonstrate that ML methods outperform linear models in predicting equity returns from fundamental predictors. By extension, if ESG ratings contain a systematic component anchored in firm fundamentals — such as carbon intensity, board structure, and ownership concentration — ML models should recover this component more effectively than OLS or Ridge regression. A meaningful R^2 would validate the premise that ESG ratings are at least partially explained by observable fundamentals, rather than being entirely idiosyncratic to each rating agency.

Hypothesis 2 — Characteristics of High-Inconsistency Firms

H2₀: The ESG inconsistency score (observed ESG rating minus ML-predicted rating) is not systematically related to firm disclosure intensity, size, or sector.

H2₁: Firms with high positive ESG inconsistency scores (rated above what their fundamentals justify) exhibit significantly higher disclosure intensity, larger firm size, and higher ESG rating disagreement across providers, consistent with appearance-driven reporting.

Rationale: Christensen, Hail, and Leuz (2021) show that mandatory sustainability disclosure does not always correspond to changes in real firm behavior. Drempetic, Klein, and Zwergel (2020) document that larger firms receive systematically higher ESG scores, partly because they have more resources for reporting. If high

inconsistency is systematically clustered among firms with high disclosure intensity and large size, this is consistent with ESG ratings being inflated by communication quality rather than by genuine sustainability improvements. This hypothesis also connects to Gibson Brandon, Krueger, and Schmidt (2021), who show that cross-provider ESG disagreement is economically meaningful.

Hypothesis 3 — Portfolio Performance of Inconsistency-Adjusted ESG

H3₀: ESG-tilted portfolios constructed using inconsistency-adjusted ESG scores do not outperform portfolios using raw ESG scores on a risk-adjusted basis.

H3₁: ESG-tilted portfolios constructed using inconsistency-adjusted ESG scores achieve higher out-of-sample Sharpe ratios and lower maximum drawdowns than portfolios using raw ESG scores.

Rationale: Pedersen, Fitzgibbons, and Pomorski (2021) model the ESG-efficient frontier and show that investors who have better information about ESG quality can achieve improved risk-return outcomes. By filtering out the noise component of ESG ratings through the inconsistency correction, the adjusted signal should be more closely aligned with economically meaningful sustainability risks — particularly those related to climate transition risk (Bolton and Kacperczyk, 2021) and governance quality. This should translate into better portfolio composition and improved out-of-sample performance, as found in related contexts by Friede, Busch, and Bassen (2015) in their large-scale meta-analysis of ESG-performance studies.

Hypothesis 4 (Auxiliary) — Signal Purity

H4₀: The inconsistency-adjusted ESG score is not less correlated with cross-provider ESG rating disagreement than the raw ESG score.

H4₁: The inconsistency-adjusted ESG score exhibits lower cross-provider rating disagreement, as measured by the standard deviation of ESG scores across providers for the same firm, compared to the raw ESG score.

Rationale: If the inconsistency correction successfully removes the disclosure-driven noise from ESG ratings, the resulting signal should be more consistent across providers, since providers differ less in their assessment of hard fundamentals

(emissions data, governance structures) than in their weighting of qualitative disclosure elements (Berg, Kölbel, and Rigobon, 2022). Testing this auxiliary hypothesis allows for an additional validation of the inconsistency framework independent of portfolio outcomes.

4. Literature Review

The literature review is organized into five thematic areas: (i) the construction and divergence of ESG ratings, (ii) ESG integration and financial performance, (iii) greenwashing and disclosure incentives, (iv) machine learning applications in finance and ESG, and (v) portfolio construction under ESG constraints. Together, these strands motivate the methodology proposed in Section 5.

4.1 ESG Ratings: Construction, Divergence, and Disagreement

ESG ratings have emerged as the dominant instrument through which institutional investors assess the non-financial performance of publicly listed firms. However, these ratings are produced by a fragmented and proprietary industry in which major providers — including MSCI, Sustainalytics, Refinitiv (LSEG), and S&P Global — apply distinct methodological frameworks, weighting schemes, and data collection practices. The consequences of this heterogeneity are substantial.

Berg, Kölbel, and Rigobon (2022) provide the most comprehensive analysis of ESG rating divergence to date. Using a decomposition framework applied to ratings from six major providers, they attribute disagreement to three sources: scope disagreement (providers measuring different attributes), measurement disagreement (providers measuring the same attribute with different indicators), and weight disagreement (providers assigning different importance to the same attribute). Crucially, they show that measurement disagreement is the dominant driver, suggesting that the problem is not merely one of emphasis but of fundamentally different assessments of the same underlying firm characteristics. The average pairwise correlation between provider ratings in their sample is approximately 0.54 — far lower than the near-perfect correlation observed across credit ratings from Moody's and S&P (Berg et al., 2022).

Gibson Brandon, Krueger, and Schmidt (2021) build on this insight by documenting that ESG rating disagreement is not only a statistical curiosity but is economically meaningful. They show that firms in the top quartile of ESG disagreement earn average annual returns approximately 1.5% higher than firms in the bottom quartile — consistent with ESG uncertainty being priced by investors who demand a risk premium for ambiguous sustainability signals. This finding has direct implications for portfolio

construction: portfolios built on noisy ESG signals may systematically misprice ESG-related risk.

Kotsantonis and Serafeim (2019) complement this evidence by identifying four structural problems in ESG data: inconsistency in the definition of metrics across firms and years, errors in raw data due to reliance on voluntary disclosure, the absence of third-party verification for most ESG data points, and the difficulty of benchmarking across sectors with structurally different ESG exposures. Their work emphasizes that the challenge is not merely one of weighting but of the fundamental reliability of the underlying data inputs.

Drempetic, Klein, and Zwergel (2020) investigate a related but underexplored dimension: the systematic influence of firm size on ESG scores. Their analysis of over 3,000 firms finds that larger firms receive significantly higher ESG scores, even after controlling for actual sustainability performance. They attribute this to the information production advantage of large firms, which employ dedicated sustainability departments and reporting specialists. This finding motivates the inclusion of firm size as a control variable in the proposed ML models, as failure to account for this effect would confound the inconsistency score.

4.2 ESG Integration and Financial Performance

A vast empirical literature has examined whether ESG integration generates a financial premium or imposes a cost. Friede, Busch, and Bassen (2015) conduct the most comprehensive meta-analysis of this literature, synthesizing over 2,000 empirical studies. They find that approximately 90% of studies report a non-negative relationship between ESG criteria and corporate financial performance, with the majority reporting a positive association. However, this aggregate finding masks considerable heterogeneity across regions, sectors, ESG dimensions, and time periods.

In an asset pricing framework, Pástor, Stambaugh, and Taylor (2021) formalize the conditions under which ESG preferences generate a financial premium or discount. In their equilibrium model, stocks with high ESG characteristics earn lower expected returns in steady state because ESG-conscious investors are willing to hold them at a premium, accepting lower expected returns in exchange for non-pecuniary utility. This prediction implies that a naive long-only ESG tilt should not generate alpha on a risk-

adjusted basis. However, they also show that if ESG concerns are rising over time — as has been empirically the case — ESG stocks may earn positive realized returns in the transition period, a finding consistent with the strong ESG performance observed in the 2010s.

Pedersen, Fitzgibbons, and Pomorski (2021) develop the concept of the ESG-efficient frontier: the mean-variance frontier available to investors who differ in their degree of ESG awareness. Crucially, they show that an investor who has better information about ESG quality — specifically, who can distinguish firms with genuinely superior ESG performance from firms that merely appear sustainable — can achieve a dominant portfolio. This result directly motivates the present thesis: if the ML-based inconsistency correction recovers a cleaner ESG signal, investors using it should operate on a frontier that dominates those relying on raw, noisy ESG scores.

Bolton and Kacperczyk (2021) focus on the environmental dimension and provide evidence that carbon risk is priced in equity markets. Firms with higher total carbon emissions earn higher returns, consistent with investors demanding compensation for exposure to climate transition risk. Their findings establish that at least some ESG-relevant information — specifically, physical emissions data — is both observable and priced. This supports the thesis's use of carbon emissions intensity as a primary ESG fundamental, as it is both economically meaningful and largely independent of voluntary disclosure choices.

Lins, Servaes, and Tamayo (2017) provide complementary evidence on the social dimension. Using the 2008-2009 financial crisis as a natural experiment, they show that firms with high social capital — proxied by CSR activities — significantly outperformed their peers during the crisis, suggesting that trust built through genuine stakeholder engagement provides insurance value during periods of market stress. This result underscores the relevance of ESG integration beyond narrow financial optimization and motivates the inclusion of controversy indicators as social fundamentals in the proposed model.

4.3 Greenwashing, Disclosure Incentives, and ESG Inconsistency

A parallel literature examines the strategic dimension of ESG rating management, raising concerns about the reliability of rating agencies' assessments as signals of true sustainability performance.

Christensen, Hail, and Leuz (2021) analyze the consequences of mandatory CSR and sustainability reporting regulations, examining how disclosure mandates affect both the quantity and quality of ESG-related information. Their central finding is that mandatory disclosure does not necessarily improve the informativeness of sustainability reporting; instead, it often leads firms to produce standardized, compliance-oriented reports that satisfy formal requirements without meaningfully increasing transparency about real economic impacts. They identify a fundamental tension between disclosure quantity — which tends to increase under mandates — and disclosure quality, which may deteriorate if firms learn to manage the content of their reports strategically.

This phenomenon — broadly captured by the concept of greenwashing — has received growing empirical attention. When firms can improve ESG ratings through enhanced communication, investor relations efforts, or strategic topic selection in sustainability reports without changing underlying operations, the informational content of ESG scores is directly compromised. The implication for portfolio construction is that portfolios built on ESG scores contaminated by greenwashing may systematically overweight firms with superior communication skills rather than superior sustainability practices.

The present thesis operationalizes this concern through the concept of ESG inconsistency: the gap between a firm's observed ESG rating and the rating that would be predicted from its observable fundamentals alone. A firm with a large positive inconsistency score — rated higher than its fundamentals would justify — may be benefiting from superior disclosure quality, greenwashing, or favorable treatment by rating agencies due to disclosure volume. By quantifying this gap using machine learning, the thesis provides a novel instrument for distinguishing appearance from substance in ESG ratings.

Riedl and Smeets (2017) provide a complementary behavioral perspective by examining why investors hold socially responsible mutual funds despite their lower risk-adjusted returns in many periods. Their evidence that social preferences and

identity motivate ESG investment decisions — independently of financial expectations — implies that ESG-driven demand is not fully arbitrage-able, which in turn allows informational distortions in ESG ratings to persist without being immediately corrected by the market.

4.4 Machine Learning in Finance and ESG Applications

The application of machine learning to financial prediction problems has expanded rapidly since the seminal contribution of Gu, Kelly, and Xiu (2020), who evaluate a comprehensive set of ML methods — including elastic net, random forests, gradient boosted trees, and neural networks — for the prediction of monthly equity returns using a large panel of firm characteristics. Their key findings are that non-linear ML methods, particularly gradient boosted trees and neural networks, substantially outperform linear benchmarks in out-of-sample prediction, and that the gains are concentrated in the ability of ML to capture interaction effects among predictors.

These methodological insights are directly transferable to the ESG context. The relationship between firm fundamentals and ESG ratings is unlikely to be purely linear: the marginal effect of an additional unit of carbon emissions on an ESG score may depend on sector, firm size, and existing governance structure. Random forests and gradient boosting algorithms — specifically XGBoost (Chen and Guestrin, 2016) and LightGBM — are well suited to capturing these non-linearities while remaining interpretable through feature importance measures and SHAP (SHapley Additive exPlanations) values (Lundberg and Lee, 2017).

Breiman (2001) introduced random forests as an ensemble method that reduces variance through bootstrap aggregation (bagging) and random feature selection. For tabular financial data, random forests have proven highly competitive with more complex models, offering strong predictive accuracy alongside native feature importance measures. Their robustness to outliers and missing values makes them particularly suitable for ESG data, which is known to suffer from measurement errors and gaps in disclosure (Kotsantonis and Serafeim, 2019).

Chen and Guestrin (2016) developed XGBoost as a scalable gradient boosting framework that combines the predictive power of tree ensembles with computational efficiency and built-in regularization. XGBoost has become a standard benchmark in

applied financial machine learning due to its consistent outperformance on tabular datasets and its interpretability through SHAP-based decompositions. The proposed methodology will implement XGBoost as the primary ML model, with a Ridge regression baseline and Random Forest for comparison.

Lundberg and Lee (2017) proposed SHAP values as a theoretically grounded method for interpreting the predictions of any ML model by attributing each prediction to its input features in a manner consistent with Shapley values from cooperative game theory. In the ESG context, SHAP analysis will allow us to quantify the contribution of each ESG fundamental — carbon intensity, board independence, CEO duality, controversy scores, and financial controls — to the predicted ESG rating for each firm. This provides an economically interpretable decomposition of the ML predictions and allows for the identification of the specific fundamentals that most drive inconsistency.

4.5 Portfolio Construction under ESG Constraints

The portfolio construction literature has developed several approaches to integrating ESG signals into mean-variance optimization. The most common approaches are ESG-tilted portfolios — which overweight high-ESG firms relative to a benchmark — and hard-constraint portfolios — which exclude firms below a minimum ESG threshold. Both approaches are standard in the industry and have been studied extensively in the academic literature.

Dimson, Marsh, and Staunton (2020) examine the long-run historical performance of sustainable investing strategies across multiple countries and time periods. Their evidence suggests that the financial performance of ESG portfolios depends critically on the quality of the ESG signal used: portfolios constructed from fundamentals-based ESG proxies (such as governance indicators or carbon data) tend to show more stable performance than those built from composite ESG scores, which are noisier.

The proposed portfolio evaluation framework follows standard practice in the empirical asset pricing literature. Portfolio performance will be assessed out of sample using returns, volatility, Sharpe ratios, maximum drawdown, and turnover. Consistent with Pástor, Stambaugh, and Taylor (2021), the analysis does not seek to establish that ESG investing systematically outperforms conventional strategies on a risk-adjusted basis — which equilibrium theory suggests should not hold unconditionally. Instead,

the focus is on the relative performance of the inconsistency-adjusted ESG signal versus the raw ESG signal, controlling for all other portfolio construction choices.

This comparative design is critical for isolating the informational contribution of the inconsistency correction. By holding constant the portfolio construction methodology — ESG-tilt weights, rebalancing frequency, and the investment universe — and varying only the ESG signal (raw versus adjusted), any difference in portfolio outcomes can be attributed to the quality of the ESG measurement rather than to differences in portfolio construction strategy.

5. Proposed Methodology

The empirical methodology proceeds in four sequential steps: (i) data construction and feature engineering, (ii) ML model training and selection, (iii) ESG inconsistency score construction, and (iv) portfolio construction and out-of-sample evaluation.

5.1 Data Construction and Feature Engineering

The analysis will be conducted on a panel of publicly listed firms over the period 2015–2023, encompassing at least 200 firms per year across multiple sectors. ESG ratings will be obtained from Yahoo Finance via the `yfinance` Python library, which provides Sustainalytics-based ESG scores at the firm level. Financial and accounting fundamentals will be drawn from publicly available sources including Yahoo Finance annual reports data, SEC EDGAR for governance variables, and the Carbon Disclosure Project (CDP) for carbon emissions data where available. Where university access to WRDS/Compustat is available, this will be used as the primary source for financial statement data.

ESG fundamentals are defined as firm-level observable characteristics that capture real environmental, social, and governance exposures, independently of how a firm chooses to communicate about them. The feature set is organized into three categories:

Environmental Fundamentals:

- Carbon emissions intensity (Scope 1 and Scope 2 emissions per unit of revenue)
- Year-on-year change in emissions intensity (trend)
- Energy consumption intensity

Governance Fundamentals:

- Board independence ratio (fraction of independent directors)
- CEO-Chair duality (binary indicator)
- Institutional ownership concentration
- Audit committee independence

Social and Controversy Indicators:

- ESG controversy score (capturing labor, human rights, and product safety incidents)
- Number of controversies in the trailing twelve months

Financial Control Variables:

- Market capitalization (log-transformed)
- Return on assets (profitability)
- Leverage ratio (total debt / total assets)
- Sector fixed effects (GICS level 1)

All continuous variables will be standardized (zero mean, unit variance) within each year-sector group to remove systematic cross-sectional and time-series variation unrelated to ESG fundamentals. Missing values will be addressed through median imputation within industry-year groups, consistent with best practice in financial panel data analysis.

5.2 Machine Learning Model Training

Three ML models will be trained to predict ESG ratings from the feature set described above. Model training follows a time-series cross-validation structure (expanding window), where the model is trained on all data up to year t and evaluated on year $t+1$. This approach prevents look-ahead bias and ensures that all reported performance metrics are genuinely out of sample.

Model 1 — Ridge Regression (Baseline):

Ridge regression provides a penalized linear baseline that controls for overfitting while remaining fully interpretable. The regularization parameter λ is selected via cross-validation. Ridge regression represents the counterfactual: how well can a linear model predict ESG ratings from fundamentals?

Model 2 — Random Forest:

A random forest ensemble of 500 decision trees will be trained, with maximum depth and minimum samples per leaf tuned via grid search. Random forests naturally handle non-linear relationships and interactions between features, and provide native feature

importance measures based on impurity reduction. Their robustness to outliers makes them particularly suitable for ESG data.

Model 3 — XGBoost (Primary Model):

XGBoost will serve as the primary predictive model, given its consistent state-of-the-art performance on tabular financial data (Gu, Kelly, and Xiu, 2020; Chen and Guestrin, 2016). Hyperparameters — learning rate, maximum depth, number of estimators, and L1/L2 regularization — will be tuned via Bayesian optimization with five-fold time-series cross-validation. SHAP values (Lundberg and Lee, 2017) will be computed to provide firm-level, feature-level attribution of predictions.

Model selection will be based on out-of-sample R^2 and root mean squared error (RMSE). The best-performing model will be used to generate the ESG inconsistency scores, with the other models used for robustness checks.

5.3 ESG Inconsistency Score Construction

For each firm i in year t , the ESG inconsistency score is defined as:

$$\text{Inconsistency}_{i,t} = \text{ESG_Observed}_{i,t} - \text{ESG_Predicted}_{i,t}$$

where ESG_Observed is the raw ESG score from the rating provider and ESG_Predicted is the out-of-sample prediction from the best-performing ML model. A positive inconsistency score indicates that a firm's ESG rating is higher than what its observable fundamentals would predict — suggesting that the rating may reflect superior disclosure quality, strategic communication, or rating inflation rather than genuine sustainability performance. A negative inconsistency score indicates a potentially underrated firm.

The inconsistency score is standardized within each year (zero mean, unit standard deviation) to remove time-series trends and facilitate comparisons across periods. Firms are then sorted into quintiles based on their inconsistency score, allowing for an analysis of the firm characteristics that are systematically associated with high versus low inconsistency (addressing H2).

The adjusted ESG score is constructed by penalizing the raw ESG score for inconsistency:

$$\text{ESG_Adjusted}_{it} = \text{ESG_Observed}_{it} - \alpha \times \text{Inconsistency}_{it}$$

where α is a penalty parameter calibrated to optimize the signal-to-noise ratio in the training period. The baseline analysis uses $\alpha = 1$, which fully subtracts the inconsistency. Robustness checks will examine $\alpha \in \{0.5, 1.0, 1.5, 2.0\}$ to assess the sensitivity of portfolio outcomes to the penalty intensity.

5.4 Portfolio Construction and Evaluation

ESG-tilted portfolios are constructed using a standard z-score tilt methodology. The portfolio weight of firm i at time t is:

$$w_{it} = w_{it}^{\text{INDLx}} + \gamma \times z(\text{ESG}_{it}) / N_t$$

where w_{it}^{INDLx} is the benchmark weight (equal-weighted or market cap-weighted), $z(\text{ESG}_{it})$ is the standardized ESG score (raw or adjusted), γ controls tilt intensity, and N_t is the number of firms in the investment universe. This approach is standard in the quantitative ESG portfolio literature and ensures that differences in portfolio performance are attributable to the ESG signal rather than to differences in diversification or turnover.

Three portfolios will be constructed and compared: (1) Benchmark Portfolio (equal-weighted, no ESG tilt); (2) Raw ESG Portfolio (ESG tilt based on raw Sustainalytics-based ESG scores); and (3) Adjusted ESG Portfolio (ESG tilt based on inconsistency-adjusted ESG scores).

Portfolio performance will be evaluated on an annual rebalancing cycle over the period 2018–2023, using the first three years (2015–2017) as the training window for the initial ML model. Performance metrics include: annualized return, annualized volatility, Sharpe ratio, maximum drawdown, Calmar ratio, and portfolio turnover. Statistical significance of performance differences will be assessed using the Jobson-Korkie test

for differences in Sharpe ratios (Jobson and Korkie, 1981), with bootstrap confidence intervals.

6. Experimental Data

Given the constraints on proprietary data access, the empirical analysis will rely on a combination of freely available and academically accessible data sources. The investment universe will consist of firms included in major public indices (S&P 500 or MSCI World constituents), for which ESG coverage is highest among freely available providers.

6.1 Primary Data Sources

ESG Ratings:

ESG scores will be sourced from Yahoo Finance via the `yfinance` Python library. Yahoo Finance provides Sustainalytics-based ESG risk ratings at the firm level, covering Environmental, Social, and Governance sub-scores as well as an aggregate ESG risk score. While a single-provider dataset limits the ability to directly test H4 (signal purity across providers), robustness checks will use any additional freely available ESG data, including ESG Book's open dataset for European firms and sector-specific sustainability disclosures.

Stock Market Data:

Stock price data, market capitalization, and trading volume will be downloaded from Yahoo Finance via `yfinance`. Monthly return series will be computed from adjusted closing prices. This data is freely available for the full sample period (2015–2023).

Financial Fundamentals:

Financial statement data (total assets, revenues, net income, total debt, operating cash flow) will be obtained from Yahoo Finance annual financials, supplemented by SEC EDGAR (XBRL structured filings) for US-listed firms. Where university access to WRDS/Compustat is confirmed, this will serve as the primary source for financial statement data, given its superior coverage and data quality.

Governance Data:

Board composition variables (board independence ratio, CEO-Chair duality) will be hand-collected from annual proxy statements (DEF 14A filings) available on SEC EDGAR for US-listed firms, and from company annual reports for non-US firms. This approach is consistent with academic practice when proprietary governance databases (BoardEx, ISS) are unavailable.

Carbon Emissions Data:

Carbon emissions data (Scope 1 and Scope 2) will be obtained from company sustainability reports and the Carbon Disclosure Project (CDP) public database, which provides self-reported emissions data for CDP signatories. CDP data is freely downloadable upon registration for academic research purposes. For firms not reporting to CDP, emissions intensity will be estimated using sector-level emission factors as a fallback.

6.2 Sample Construction

The target sample consists of S&P 500 constituent firms over the period 2015–2023, providing a minimum of 400 firm-year observations with complete ESG and fundamental data. The sample will be restricted to firm-year observations where both the ESG score and at least 70% of the fundamental variables are available. Survivorship bias will be mitigated by including firms that were delisted during the sample period up to their last available observation.

The panel structure allows for the use of time-series cross-validation in the ML training step, with an expanding window that ensures no future data is used in model training. Summary statistics for the final sample will be reported, including distributions of ESG scores, key fundamentals, and inconsistency scores by sector and year.

6.3 Data Limitations and Mitigations

The reliance on freely available ESG data introduces three main limitations. First, Yahoo Finance ESG scores reflect a single provider (Sustainalytics), which limits the ability to directly test H4. This will be addressed by using the cross-provider disagreement measure from Berg, Kölbel, and Rigobon (2022), which is available in replication materials, as an external benchmark. Second, the absence of a commercial

governance database introduces measurement error in governance variables. Board-level variables will be validated against a random sample of company filings to assess data quality. Third, CDP coverage is uneven across sectors and firm sizes. Missing carbon data will be treated explicitly in the analysis through sector-median imputation and a robustness check that restricts the sample to CDP reporters only.

7. Provisional Bibliography

The following references constitute the provisional bibliography for this thesis, comprising 16 academic works across the five thematic areas covered in the literature review.

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